

VFT GYM API Documentation

1. Authentication

1.1 Sign In (POST)

Endpoint: POST /api/user/

Description: Authenticates a user and returns an API token for subsequent requests.

Headers:

- Content-Type: application/json (implied)

Request Body:

```
json
{
  "email": "user1@gmail.com",
  "password": "User@123"
}
```

Response (JSON):

```
json
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

1.2 Verify Authentication (GET)

Endpoint: GET /api/user/auth

Description: Validates the current API token.

Headers:

- apitoken: {{APIToken}}

Response:

- **200 OK:** Token is valid
- **401 Unauthorized:** Token is invalid or expired

1.3 Logout (GET)

Endpoint: GET /api/user/logout

Description: Invalidates the current API token.

Headers:

- ApiToken: {{APIToken}}

Note: Header naming inconsistency - uses ApiToken instead of apitoken

2. Area Management

2.1 Get All Areas (GET)

Expected Endpoint: GET /api/area

Description: Retrieves all defined areas in the system.

Headers:

- ApiToken: {{APIToken}}

2.2 Create Area (POST)

Expected Endpoint: POST /api/area

Description: Creates a new area.

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Expected Request Body:

json

```
{  
  "AreaID": "A01",  
  "AreaName": "Area1",  
  "Description": "Default Area"  
}
```

2.3 Update Area (PUT)

Expected Endpoint: PUT /api/area/{id}

Description: Updates an existing area.

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Expected Request Body:

json

```
{  
  "AreaID": "A01",  
  "AreaName": "Area1",  
  "Description": "Default Area"  
}
```

3. Device Management

3.1 Get All Devices (GET)

Endpoint: GET /api/device

Description: Retrieves a list of all registered devices.

Headers:

- ApiToken: {{APIToken}}

Response Structure:

json

```
{
  "devices": [
    {
      "id": 1,
      "DevSN": "QEM1253800019",
      "DevName": "Dev11",
      "TimeZone": "330",
      "AreaID": 3,
    }
  ]
}
```

3.2 Add New Device (POST)

Endpoint: POST /api/device

Description: Registers a new biometric device in the system.

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Request Body:

```
json
{
  "DevSN": "QEM1253800019",
  "DevName": "Dev11",
  "TimeZone": "330",
  "AreaID": 3,
  "IsReboot": 1
}
```

Field Specifications:

Field	Type	Required	Description	Example
DevSN	String	Yes	Device serial number	"QEM1253800019"
DevName	String	Yes	Device display name	"Dev11"
TimeZone	String	Yes	Timezone offset (minutes)	"330" (UTC+5:30)
AreaID	Integer	Yes	Associated area ID	3
IsReboot	Integer	No	Auto-reboot flag (0/1)	1

3.3 Update Device (PUT)

Endpoint: PUT /api/device/{id}

Description: Updates device properties.

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Path Parameter:

- id: Device ID (integer)

Example Request:

```
json
{
  "DevName": "Dev5-Kandy(Pallekale)",
}
```

4. Device Commands

4.1 Send Device Command (POST)

Endpoint: POST /api/devicecmd

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Request Body Structure:

```
json
{
  "DevSN": "QEM1253800017",
  "Type": "General",
  "Content": "COMMAND_STRING"
}
```

Field Specifications:

Field	Type	Required	Description
DevSN	String	Yes	Target device serial number
Type	String	Yes	Command type ("General")
Content	String	Yes	ZKTeco command string

4.2 Command Catalog

User Management Commands:

javascript

// Get users with filters

"Content": "DATA QUERY tablename=user,fielddesc=*,filter=*"

// Add new user

"Content": "DATA UPDATE user

CardNo=\tPin=8\tPassword=123456\tGroup=1\tStartTime=20241010\tEndTime=20251020\t Name=John\t
privilege=0\tDisable=0"

// Update user active date range

"Content": "DATA UPDATE user Pin=8\tStartTime=20241010\tEndTime=20251020"

Biometric Template Management:

javascript

// Get fingerprint templates

"Content": "DATA QUERY tablename=templatev10,fielddesc=*,filter=*"

Access Control Commands:

javascript

// Control door 1

"Content": "CONTROL DEVICE 01010105"

Attendance & Transactions:

javascript

// Get transactions

"Content": "DATA QUERY tablename=transaction,fielddesc=*,filter=*"

Authorization Management:

javascript

// Add user authorization

"Content": "DATA UPDATE userauthorize Pin=8\tAuthorizeTimezoneld=1\tAuthorizeDoorId=1"

// Delete user authorization

"Content": "DATA DELETE userauthorize Pin=8"

5. Employee Management

5.1 Get All Employees (GET)

Expected Endpoint: GET /api/template

Description: Retrieves all gym employees/members.

Headers:

- ApiToken: {{APIToken}}

5.2 Create Employee (POST)

Expected Endpoint: POST /api/template/add

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Expected Request Body:

```
json
{
  "EmpId": "166",
  "EmpName": "Kamal"
}
```

5.3 Update Employee (PUT)

Expected Endpoint: PUT /api/template/edit/{id}

Description: Updates employee information.

Headers:

- ApiToken: {{APIToken}}
- Content-Type: application/json

Expected Request Body:

```
json
{
```

```
"EmpName": "Kamal"  
}
```

6. Integration Examples

Complete Workflow Example:

1. Authentication:

```
javascript  
// Sign in  
POST /api/user/  
{  
  "email": "user1@gmail.com",  
  "password": "User@123"  
}  
  
// Store token automatically via test script
```

2. Device Registration:

```
javascript  
// Add new device  
POST /api/device  
Headers: {  
  "ApiToken": "{{APIToken}}",  
  "Content-Type": "application/json"  
}  
  
Body: {  
  "DevSN": "QEM1253800019",  
  "DevName": "Gym Entrance",
```

```
"TimeZone": "330",  
"AreaID": 1,  
"IsReboot": 1  
}
```

3. User Management:

javascript

// Add user to device

POST /api/devicecmd

```
{  
  "DevSN": "QEM1253800017",  
  "Type": "General",  
  "Content": "DATA UPDATE user  
CardNo=\tPin=1001\tPassword=123456\tGroup=1\tStartTime=20240101\tEndTime=20250101\tName=John\tprivile  
ge=0\tDisable=0"  
}
```

// Upload fingerprint

POST /api/devicecmd

```
{  
  "DevSN": "QEM1253800017",  
  "Type": "General",  
  "Content": "DATA UPDATE userauthorize Pin=1001\tAuthorizeTimezoneId=1\tAuthorizeDoorId=1"  
}
```

4. Attendance Monitoring:

javascript

// Get today's attendance

POST /api/devicecmd

```
{
```

```
"DevSN": "QEM1253800017",  
"Type": "General",  
"Content": "DATA QUERY tablename=transaction,fielddesc=*,filter=*"  
}
```

